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Installing a Modular Sensor in GCP

This topic describes how to install a Modular Sensor in a Google Cloud Platform (GCP) environment. Refer to the following sections for details:

- **About Modular Sensors**
- **Site Preparation**
- **Configuring and Launching the Instance**
- **Applying a Token to the Installed Sensor**
- **Configuring a Static IP Address (Optional)**
- **Configuring Packet Mirroring in GCP**
- **Enabling SSSE3 for the Modular Sensor VM**

Use our example as a guideline, as you might be using a different software version.

Stellar Cyber does not support the installation of third-party software on its virtual or physical device sensors.

About Modular Sensors

Sensors provide the data gathering foundation for Stellar Cyber's OpenXDR platform, gathering the right data with context. Modular sensors are purpose-built Stellar Cyber sensors that include both the host and the Stellar Cyber monitoring software. They are provided as both physical devices (Photon sensors) and virtual machine images for different target environments.

Previous releases provided a variety of different types of device sensors, including Network, Security, and Modular. Going forward, the only type of device sensor is Modular. You can use the Modular Sensor Profile to enable whatever sensor features you like, creating the same functionality provided by the different sensor types in previous releases.

A modular sensor lets you easily add the features you like to your sensor. This helps simplify your deployment and lets you manage the VM requirements for the sensors based on the modular features they use.

Modular Sensors always include log ingestion. From there, you can enable different features as part of your modular sensor profile.

- Enable the **Network Traffic** feature to monitor the virtual environment, the physical environment if connected to the span port of a physical switch, or the LAN segment via a mirror port on a switch. The sensor monitors network and server response times and can identify applications.

  The sensor converts that information to metadata and forwards it to the DP as Interflow. The DP can then provide security, DDoS, and breach attempt detections.
- Enable the **Sandbox** and **IDS** features to improve your security posture:
  - Sandbox lets you detect malware in files and network traffic through Stellar Cyber's integrated cloud service and also provides anti-virus services.
  - IDS lets you detect intrusion attempts using both files and network traffic.



Keep in mind that VM resource requirements increase as you add more features to the Modular Sensor Profile. Refer to **Virtual Modular Sensor Provisioning and Performance** for details on the resources required to run different combinations of features in a Modular Sensor Profile, as well as how to use the **show module** and **show module request** CLI commands to compare provisioned resources against those required to run specific feature combinations. Stellar Cyber only enables a Modular Sensor Profile on a sensor if the host VM's resources can support it.

Site Preparation

Click to see the minimum system requirements for installing a modular sensor.

You must have:

- One IP address with access to a default gateway
- A Stellar Cyber license that can be applied to the sensor

To prepare for the installation:

1. Open firewall ports for log ingestion.
2. Open firewall ports for the features you plan on enabling in the Modular Sensor Profile for this sensor.
3. Contact Stellar Cyber support (support@stellarcyber.ai) for access to the image. You must provide:
   - The GCP Gmail account name

Do this at least a day before installing, so we have time to deploy the images to your account.

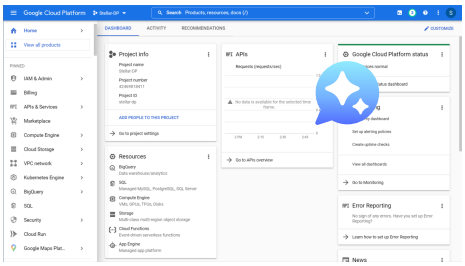
Configuring and Launching the Instance

To configure and launch the instance:

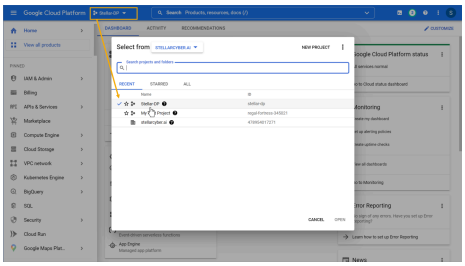


Use our example as a guideline, as you might be using a different software version.

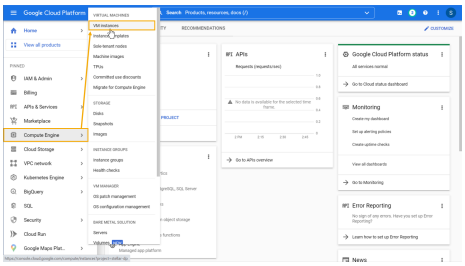
1. Log in to your **Google Cloud Platform Dashboard**.



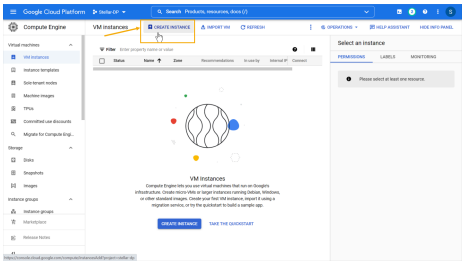
2. Use the dropdown in the toolbar to select the **Project** where you want to deploy the DP.



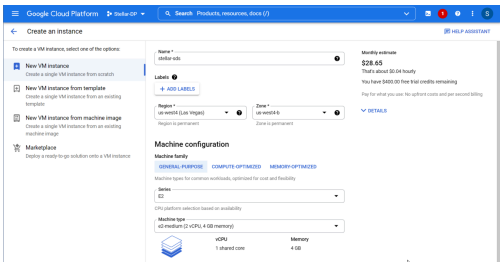
3. Select the **Compute Engine | VM instances** entry from the left navigation panel.



4. Click the **Create Instance** button.

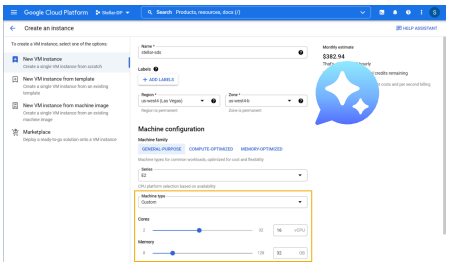


5. Supply a **Name**, **Region**, and **Zone** for the instance.

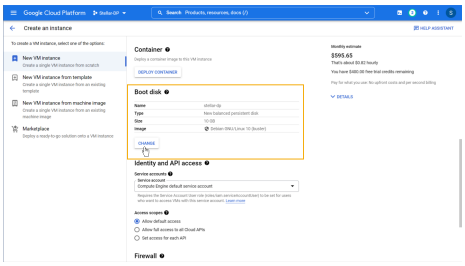


6. Choose a **Machine type** for the instance. You can either choose one of the preconfigured GCP machine types or create a **Custom** machine type, so long as the instance meets the minimum specifications.

   In the example below, we are creating a **Custom Machine Type** with 16 virtual cores and 32 GB of memory to serve as the foundation for a Security Sensor:



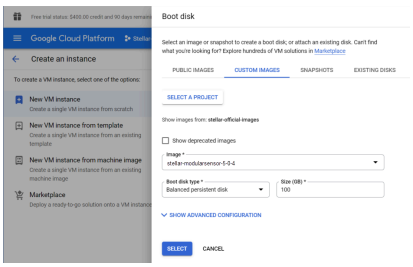
7. Scroll down to the **Boot disk** section and click the **CHANGE** button.



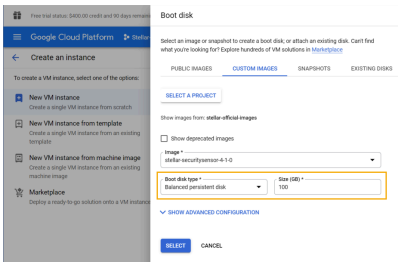
8. Click on the **CUSTOM IMAGES** tab.
9. Click the **SELECT A PROJECT** button and choose the **stellar-official-images** project.

> The **stellar-official-images** entry won't appear until we've made images available to your account. If you don't see the **stellar-official-images** entry, make sure your region is the same one where you asked Customer Success to deploy the images. If you still don't see the entry, contact Customer Success.

10. Select the entry for your sensor version and type from the **Images** dropdown.

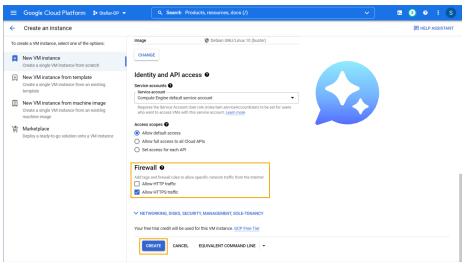


11. Set the **Boot disk type** to **Balanced persistent disk**, choose a size of **100** GB, and click **Select**.



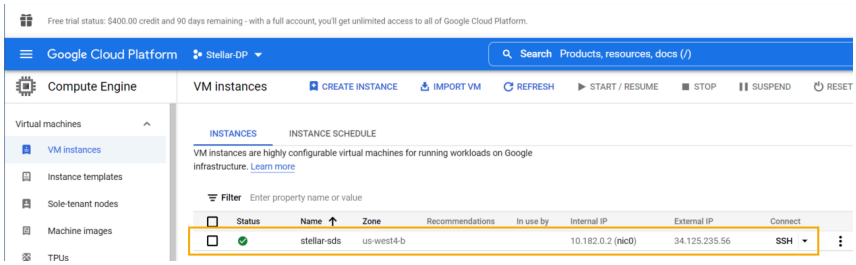
You are returned to the **Create an instance** wizard.

12. Click the **Create** button to create the instance.



You can launch the image but you cannot copy it. This means that the VM must be deployed in the GCP region where the image was authorized.

The VM is now running in the GCP cloud and appears in the **Compute engine | VM instances** list, as illustrated below. Copy the **External IP** address so you can use it to connect to the Sensor's console in the next section.



Applying a Token to the Installed Sensor

The next step is to obtain and apply the token used to authorize and configure the installed sensor.

Obtaining a Token for the Installation

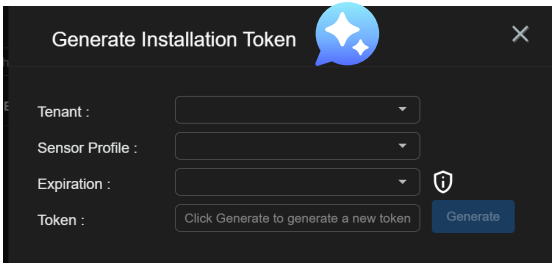
Tokens are required to authorize and configure the installation of a sensor image downloaded from the DP in the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation page**. Tokens point the installed sensor to the correct DP, assign the specified tenant, optionally provision a selected sensor profile, and authorize the sensor installation.

Use the following procedure to obtain a token in the Tokens tab:

1. Navigate to the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation** page and click on the **Tokens** tab.
2. If there is already an unexpired token that you want to use for the installation, you can either use the **Copy** button to copy it to the clipboard or use the **Download** button to download it as a file.
    - Copy the token if you plan on applying it by pasting it into a `set token string <token>` command in the CLI.
    - Download the token as a file if you plan on hosting the file on an HTTP server and referring to it in a `set token url <token url>` command.

    Refer to **Assigning Tokens** for a summary of the different ways in which tokens can be applied to a sensor installation.
3. You can also click the **Generate** button to create a new token.

The Generate Installation Token dialog appears:



4. Select the tenant for the token from the **Tenant** dropdown. This is the tenant to which all sensors authorized with this token will be automatically assigned. The dropdown lists all tenants configured for your organization in the **System | ORGANIZATION MANAGEMENT | Tenants** page.
5. Select the Sensor Profile to be assigned to all sensors authorized with this token from the **Sensor Profile** dropdown. The dropdown lists all sensor profiles available in the System | DATA SOURCE MANAGEMENT | Sensors | Sensor Profiles page for the selected Tenant.

   If you do not want to assign a Sensor Profile with a token, you can set this field to **None (no sensor profile selected)**. This is also the setting for any tokens migrated from a pre-5.3.0 release as part of an upgrade.

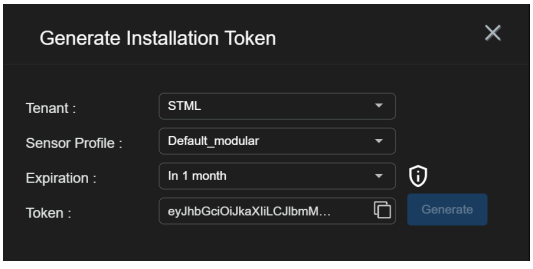
6. Use the **Expiration** dropdown to select an expiration date for this token. You can select specified expiration dates ranging from two weeks to three months.

   You can also select **Never expires** for the expiration date. However, Stellar Cyber recommends that you specify expiration dates for your tokens in order to improve the security of your deployment.

   Remember that when a token expires, **sensors authorized with the token continue to operate as normal**. Once a sensor successfully registers with the Stellar Cyber platform, it no longer uses the token. It is only used for the initial authorization, registration, and configuration of the sensor.

7. Click the **Generate** button.

   The system generates the token and displays its contents in the **Token** field. The dialog also updates to display the expiration date for the token, as illustrated below.



8. You can use the **Copy** button to copy the token to the clipboard immediately, or simply close the dialog and retrieve the token from the **Tokens** tab later on.

Applying the Token to the Sensor

Tokens are required to complete the installation of a sensor image downloaded from the DP in the **Download Image** tab.

You apply tokens to sensors as the last step in the overall installation procedure:

1. Log in to your new Sensor. The default username/password is **aella**/**changeme**. You are immediately prompted to change the password.
2. Apply the token to the installed sensor from the sensor CLI with the `set token` command using **one** of the options in the table below:

   You only need to use **one** of the options in the table below. These are just two different ways to do the same thing – apply the token.

Option 1. Copy and Paste the Token String	Copy the token string from the **Tokens** tab and paste it into the CLI command. The syntax is as follows: `set token string <pasted string>`

Option 2. Host the Token on an HTTP Server	Download the token as a file from the **Tokens** tab, upload it to an HTTP server, and reference it in the `set token` command. The syntax is as follows:
	`set token url http://<url to token>`
	You can also use an HTTPS server. In that case, the specified URL must also include the username and password for the server using the following syntax:
	`set token url https://<user:password>@URL>`

3. The CLI reports that the **Sensor token is successfully set**.

   If you receive an error message instead, it's possible that the token has expired. Refer to the **Tokens** tab to see the expiration date. If you are using the File technique, it's also possible that an extra space or line may have crept into your text file – check the file to make sure it includes only the token text.

4. Wait a minute or so. Then, verify that the token was successfully applied using any combination of the following techniques:
   - Check the **System | DATA SOURCE MANAGEMENT | Sensors | Sensors** list in the user interface to see that the sensor has registered itself successfully.
   - Verify that the `show system` command shows all services as running.
   - Verify that the `show receiver` command displays a receiver.
   - Verify that the `show json` command reports some data sent in the `BYTE_SENT` column.

Configuring a Static IP Address (Optional)

By default, the sensor uses DHCP for the management port's IP address. For ease of troubleshooting, however, Stellar Cyber recommends that you reconfigure the management port to use a static IP address. The procedure is as follows:

1. Log in to your sensor. The default username/password is **aella**/**changeme**, but you changed this when you applied the token in the previous section.
2. You can set IP parameters manually using commands similar to the following (substitute your own IP parameters for the ones shown in bold below):

   `set interface management ip` **`192.168.14.100/255.255.255.0`**

   `set interface management gateway` **`192.168.14.1`**

   `set interface management dns` **`8.8.8.8`**
3. Verify the IP settings with the `show interfaces` command.
4. Log out with the `quit` command.

Configuring Packet Mirroring in GCP

Refer to **Configuring GCP Packet Mirroring** for details on how to use packet mirroring for packet acquisition in GCP environments.

Enabling SSSE3 for the Modular Sensor VM

The Modular Sensor VM must have SSSE3 enabled for its processors in order to operate correctly. In most cases, SSSE3 will already be enabled. However, if you encounter issues with packet collection or Interflow data generation, you can use the instructions below to ensure that SSSE3 is enabled.

Click here to see instruction on how to enable SSSE3.

Sensors installed on Linux hosts must have SSSE3 enabled for their processors in order to operate correctly. This is true for Modular Sensors and Linux Server Sensors, as well as legacy Network and Security Sensors.
SSSE3 is typically supported/enabled for most vCPUs, but may not be for certain legacy AMD vCPUs. See below for instructions on enabling SSSE3.

To enable SSSE3 for a virtual machine:

1. Start the virtual shell (virsh)
2. Type the following command to edit the virtual machine's settings:

   `edit <virtual_machine_name>`
3. Locate the **<cpu>** section. It should appear similar to the following:

```
<cpu mode='custom' match='exact' check='partial'>
    <model fallback='allow'>SandyBridge</model>
......
 </cpu>
```

4. Add the following line to the **<cpu>** section:

   `<feature policy='require' name='ssse3'/>`

   When you are done, the **<cpu>** section should appear similar to the following:

```
<cpu mode='custom' match='exact' check='partial'>
    <model fallb'allow'>SandyBridge</model>
    <feature policy='require' name='ssse3'/>
.....
  </cpu>
```

5. Save changes to the virtual machine and exist virsh.
6. Run the following command:

   `virsh define /etc/libvirt/qemu/<virtual_machine_name>.xml`

   Each virtual machine has a configuration .xml file. Typically, these files are stored under **/etc/libvirt/qemu**, but the location may be different for your system.

7. Stop and start the virtual machine in virsh.

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